

Standard normal distribution

Background You are given an approximately normally distributed dataset

Task 1 Calculate the mean and standard deviation of the dataset

Task 2 Standardize the dataset

Task 3 Plot the data on a graph to see the change

Mean	Std Dev
743.03	73.49

Original dataset	Subtract Mean	Divide by std
567.45	-175.58	-2.389148399
572.45	-170.58	-2.321111376
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589.12	-153.91	-2.0943213
613.87	-129.16	-1.757538038
615.78	-127.24	-1.73145718
628.45	-114.58	-1.559096722
644.87	-98.16	-1.335708498
650.45	-92.58	-1.259733823
652.20	-90.83	-1.235920865
656.87	-86.16	-1.172419644
661.45	-81.58	-1.110052373
666.45	-76.58	-1.04201535
667.70	-75.33	-1.025006095
668.95	-74.08	-1.007996839
675.28	-67.74	-0.92181661
675.78	-67.24	-0.915012908
685.53	-57.49	-0.782340714
694.28	-48.74	-0.663275924
697.62	-45.41	-0.617917909
705.78	-37.24	-0.506790772
705.87	-37.16	-0.505656822
708.12	-34.91	-0.475040162
711.03	-31.99	-0.435351899
714.03	-28.99	-0.394529685
716.03	-26.99	-0.367314876
722.28	-20.74	-0.282268598
728.12	-14.91	-0.202892071
728.70	-14.33	-0.194954419
729.03	-13.99	-0.190418617
730.12	-12.91	-0.175677262
731.95	-11.08	-0.150730354
735.03	-7.99	-0.10877419
736.95	-6.08	-0.082693331
737.37	-5.66	-0.077023579
738.28	-4.74	-0.064550125
739.78	-3.24	-0.044139018
740.62	-2.41	-0.032799515

Mean	Std Dev
0.00	1.00

Standardising the graph results in very little difference

However, the mean is now 0 and the standard deviation 1

The original histogram has had its mean move to 0 and the sides squashed to give a standard deviation of 1

